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Fall Semester 2025-26

Lab Assignment – 5

Slot: L13+L14

Class: VL2025260105679

Branch: B.tech CSBS

Course code & title: CBS3005

Cloud, Microservices and Applications

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22BBS0105

Task 2: Analyze API Activities using AWS CloudTrail

Description:
Enable AWS CloudTrail to monitor and log all API activities in the AWS account. Also, analyze events to detect unauthorized or unusual access.

Steps:

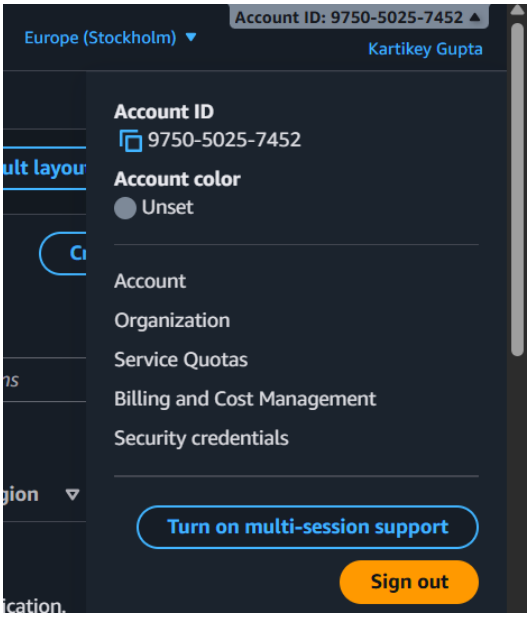
1. Go to CloudTrail Console → Trails → Create Trail.
2. Choose:
    - Trail name: MySecurityTrail
    - Apply trail to All regions
    - Store logs in a new S3 bucket
    - Enable CloudWatch Logs Integration (optional).
3. Perform a few activities in your AWS account:
    - Launch or stop an EC2 instance.
    - Create an S3 bucket.
4. Return to CloudTrail → Event History and view recent events.
5. Identify:
    - Who performed the action (IAM user or role)
    - Time of action
    - Source IP address
    - Affected resource

Expected Output:

- Screenshot of CloudTrail event list.
- Sample JSON event record for one operation.
- Short explanation of how CloudTrail supports auditing and incident investigation.

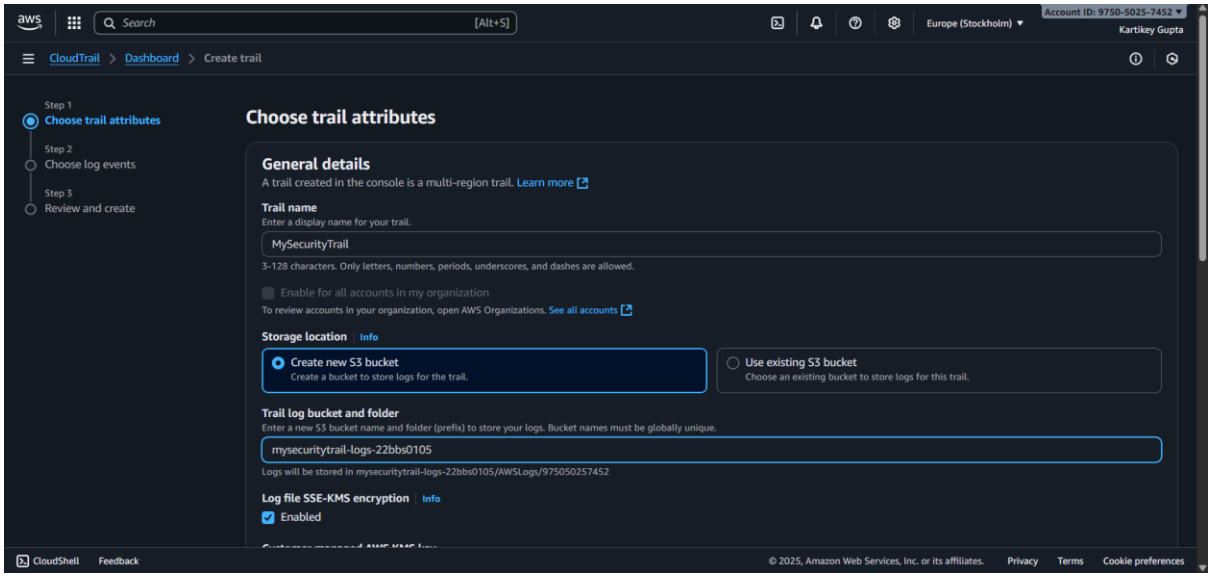
Step 1: Log in to the AWS Management Console

- Sign in to your AWS account with **Administrator access**.
- In the search bar, type **"CloudTrail"** and open the **CloudTrail** service.

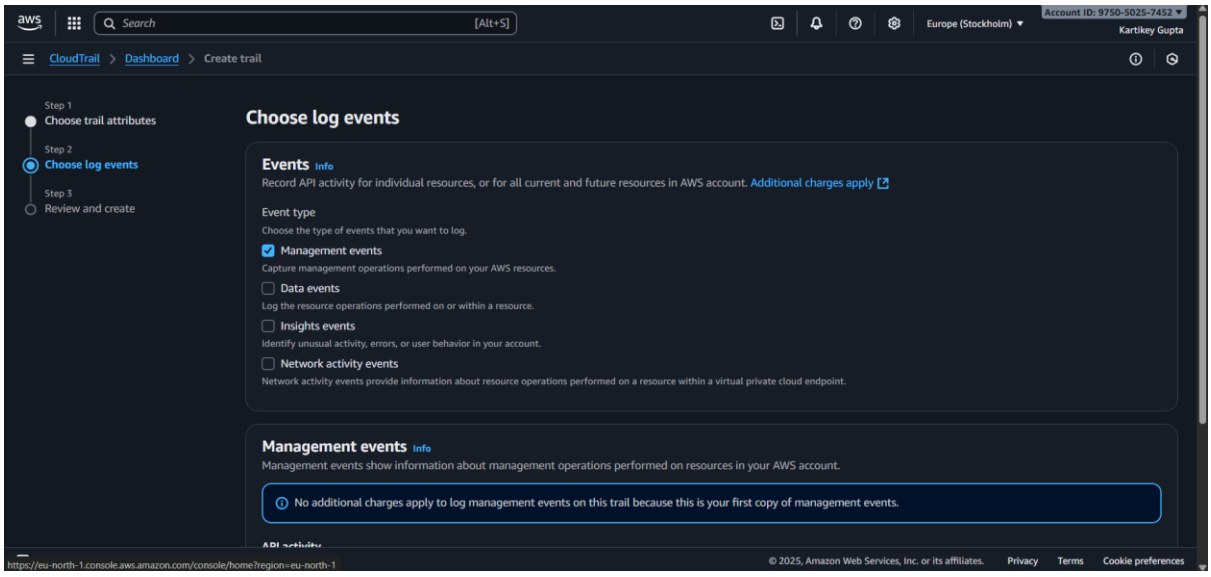


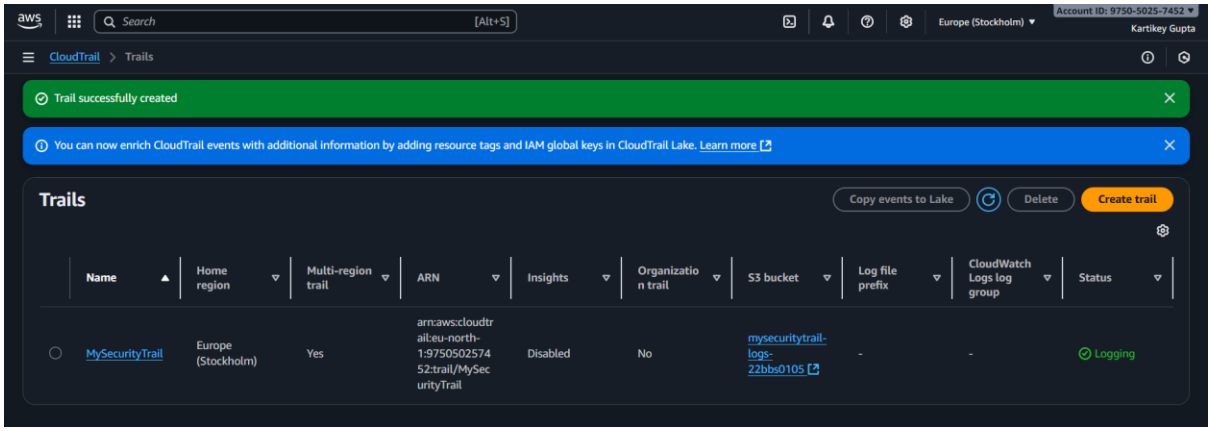
Step 2: Create a New Trail

1. In the **CloudTrail Console**, go to **Trails** → click **Create trail**.
2. Enter details:
    - **Trail name:** MySecurityTrail
    - **Apply trail to all regions: Yes**
    - **Storage location:**
        - Choose **Create a new S3 bucket**.
        - Enter a unique bucket name (mysecuritytrail-logs-22bbs0105).



3. Click **Next**, review the settings, and select **Create trail**.

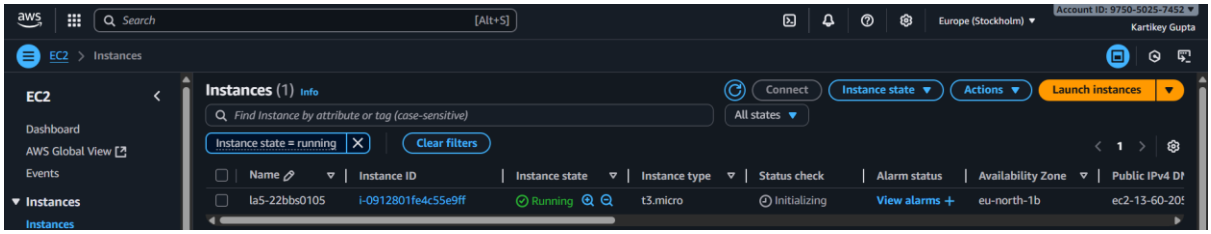




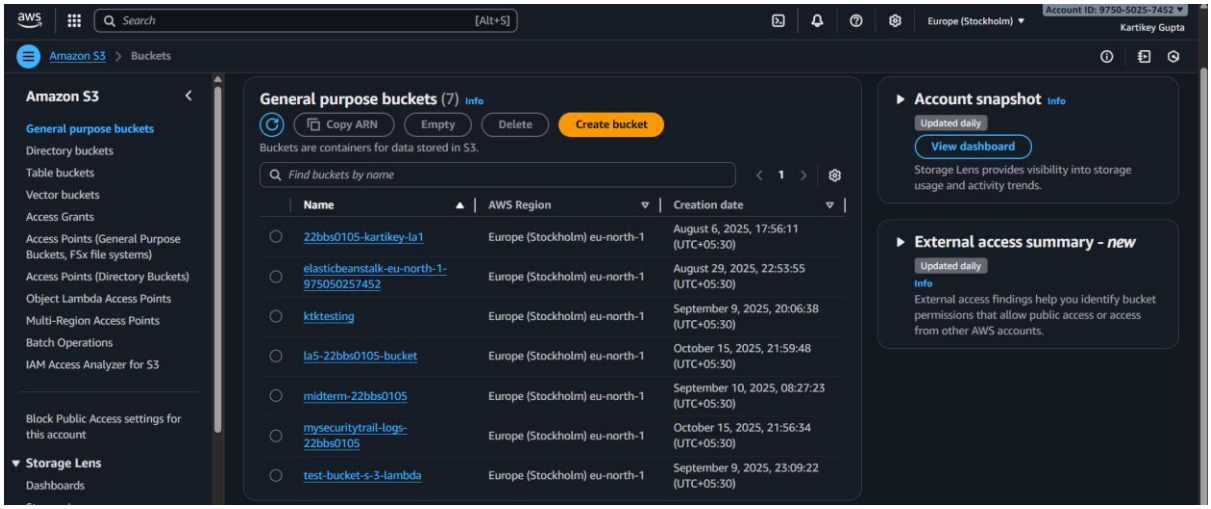
Step 3: Perform AWS Activities to Generate Logs

Perform a few actions to generate CloudTrail events:

- Go to the **EC2 Console** → **Launch Instance**



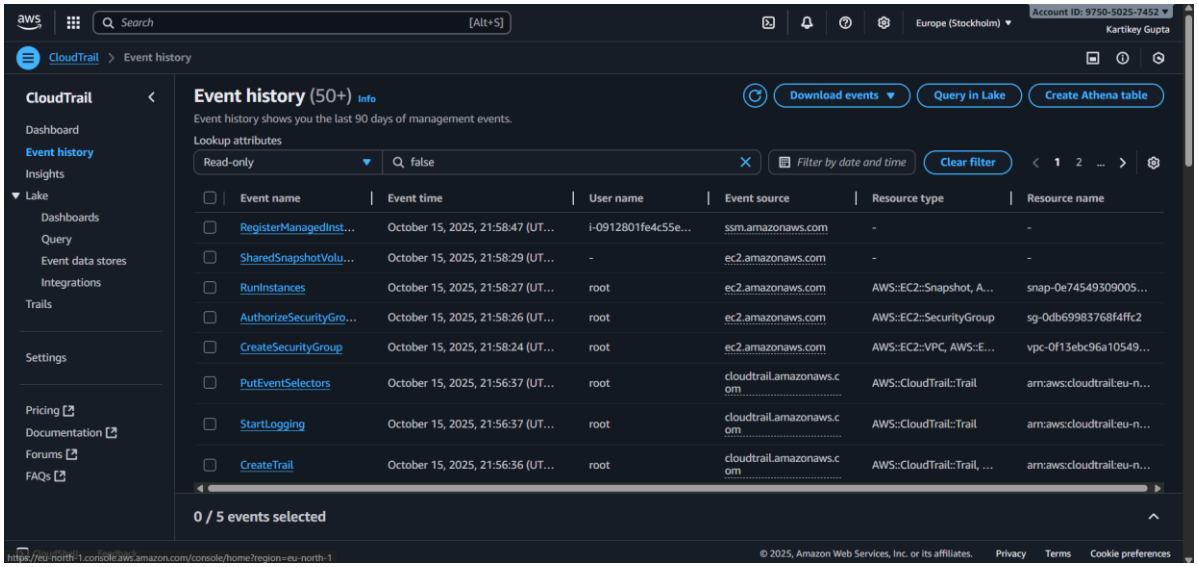
- Go to the **S3 Console** → **Create a new S3 bucket**.



Wait 2–5 minutes after performing these actions — CloudTrail may take some time to log them.

Step 4: View CloudTrail Events

1. Return to the **CloudTrail Console** → Click **Event History** (left menu).
2. You'll see a list of recorded events.

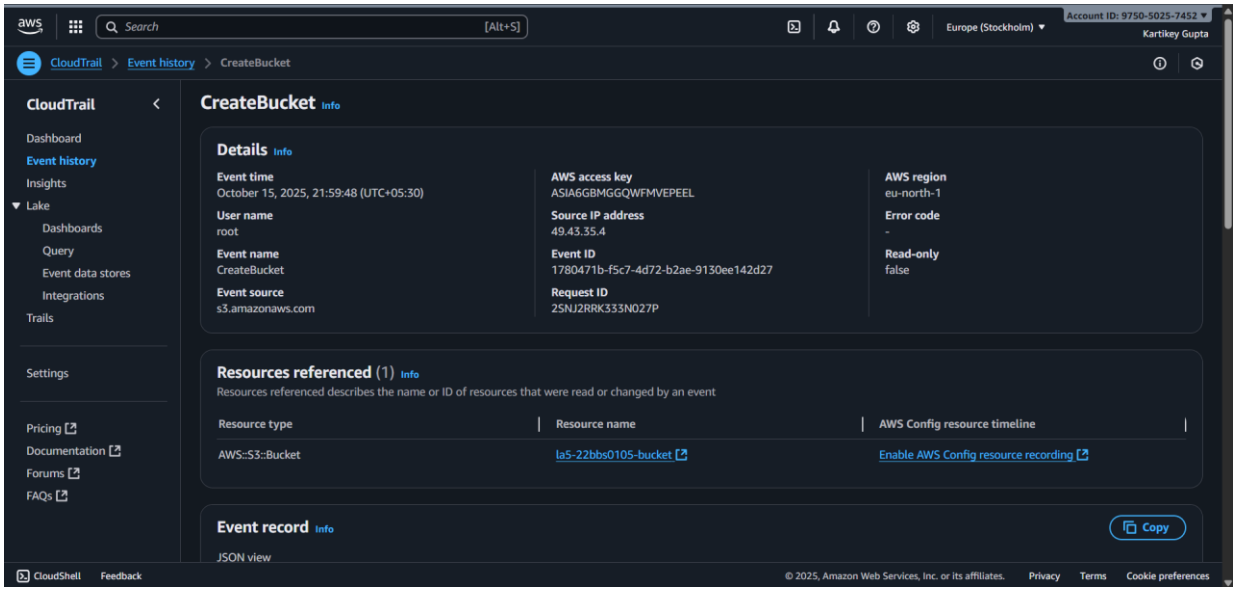


3. Use **filters** to narrow down results:
    - **Event Name:** RunInstances, StopInstances, CreateBucket, etc.
    - **Event Source:** e.g., ec2.amazonaws.com, s3.amazonaws.com.

Step 5: Identify Key Details

Click on any event to open its details.
From the JSON record, note the following:

- **Who performed the action:** (Look for "userIdentity": {"userName": ...})
- **Time of action:** "eventTime"
- **Source IP address:** "sourceIPAddress"
- **Affected resource:** "resources" or "requestParameters"



Step 6: Export Required Output

- Take a screenshot of your **CloudTrail Event History** page showing recorded activities.
- Open one event → click **View Event** → copy the JSON content.


```
{
  "eventVersion": "1.11",
  "userIdentity": {
    "type": "Root",
    "principalId": "975050257452",
    "arn": "arn:aws:iam::975050257452:root",
    "accountId": "975050257452",
    "accessKeyId": "ASIA6GBMGGQWFMVEPEEL",
    "sessionContext": {
      "attributes": {
        "creationDate": "2025-10-15T16:18:12Z",
        "mfaAuthenticated": "true"
      }
    }
  },
  "eventTime": "2025-10-15T16:29:48Z",
  "eventSource": "s3.amazonaws.com",
  "eventName": "CreateBucket",
  "awsRegion": "eu-north-1",
  "sourceIPAddress": "49.43.35.4",
  "userAgent": "[Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/140.0.0.0 Safari/537.36]",
  "requestParameters": {
    "CreateBucketConfiguration": {
      "LocationConstraint": "eu-north-1",
      "xmlns": "http://s3.amazonaws.com/doc/2006-03-01/"
    },
    "bucketName": "la5-22bbs0105-bucket",
```

```
        "Host": "la5-22bbs0105-bucket.s3.eu-north-1.amazonaws.com"
    },
    "responseElements": null,
    "additionalEventData": {
        "SignatureVersion": "SigV4",
        "CipherSuite": "TLS_CHACHA20_POLY1305_SHA256",
        "bytesTransferredIn": 192,
        "AuthenticationMethod": "AuthHeader",
        "x-amz-id-2":
"Aj6WmmJvdKYlEw8DXu99620jKEkbadNrZB03+piEk+GpknkdSdPIetsj+B4oOnTnoiyrvwqWq8k=",
        "bytesTransferredOut": 0
    },
    "requestID": "2SNJ2RRK333N027P",
    "eventID": "1780471b-f5c7-4d72-b2ae-9130ee142d27",
    "readOnly": false,
    "eventType": "AwsApiCall",
    "managementEvent": true,
    "recipientAccountId": "975050257452",
    "eventCategory": "Management",
    "tlsDetails": {
        "tlsVersion": "TLSv1.3",
        "cipherSuite": "TLS_CHACHA20_POLY1305_SHA256",
        "clientProvidedHostHeader": "la5-22bbs0105-bucket.s3.eu-north-1.amazonaws.com"
    }
}
```

Detail	Value
Who performed the action (IAM user or role)	Root user (arn:aws:iam::975050257452:root)
Time of action	2025-10-15T16:29:48Z
Source IP address	49.43.35.4
Affected resource	S3 bucket: la5-22bbs0105-bucket

Similarly, we will have log information for ec2 instance.

Task 2: Configure CloudWatch for EC2 Instance Monitoring

Description:

Create and monitor a simple EC2 instance using **Amazon CloudWatch** , configure alarms, monitor system metrics, and analyze performance data to understand how CloudWatch helps in proactive threat and performance monitoring.

Steps:

1. Launch a **t2.micro EC2 instance** (Free Tier eligible) running Amazon Linux 2.
2. Install and configure the **CloudWatch Agent** using:

   ```
   sudo yum install amazon-cloudwatch-agent -y
   sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard
   ```

3. Choose to monitor:
   - CPU utilization
   - Memory usage
   - Disk I/O

4. Start the CloudWatch agent:

   ```
   sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
   -a fetch-config -m ec2 -c file:/opt/aws/amazon-cloudwatch-agent/bin/config.json -s
   ```

5. Go to **CloudWatch Console → Metrics → EC2** and verify data collection.

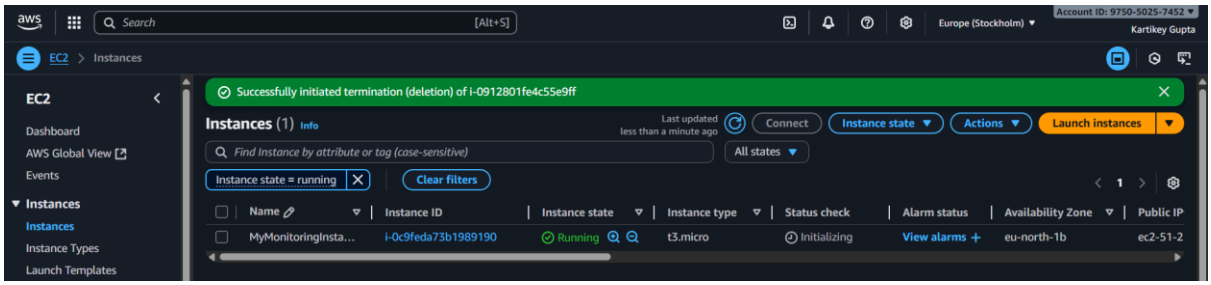
6. Create a **CloudWatch Alarm**:
   - Metric: CPUUtilization
   - Condition: >70% for 2 consecutive periods
   - Notification: via **SNS topic** (send email alert).

Expected Output:

- CloudWatch dashboard showing metrics (CPU, Memory).
- Alarm configuration screenshot.
- Email alert showing triggered event.

Step 1: Launch EC2 Instance

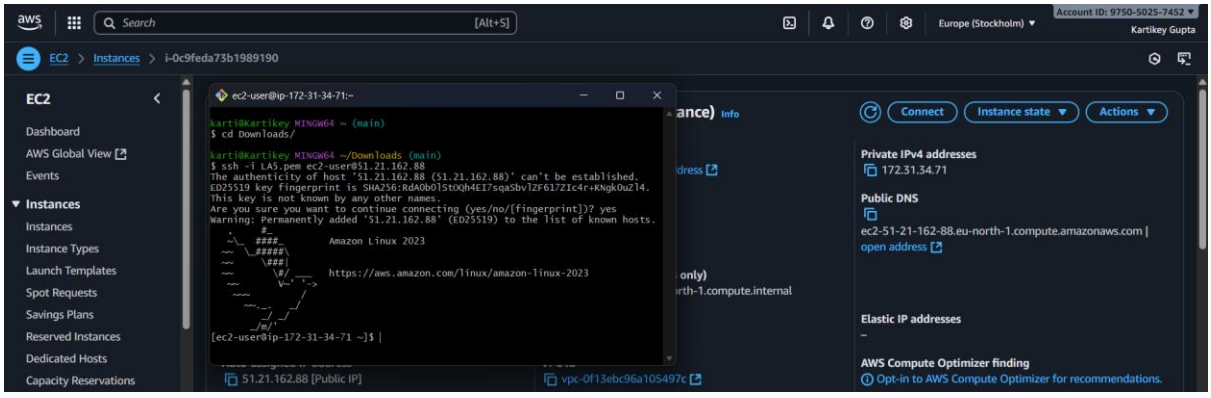
1. Go to the **AWS Management Console** → Search for **EC2** → Click **Instances → Launch instance**.
2. Configuration details:
    - **Name:** MyMonitoringInstance
    - **AMI:** Amazon Linux 2 (Free Tier eligible)
    - **Instance type:** t2.micro
    - **Key pair:** Select existing or create a new one.
    - **Security group:** Allow SSH (port 22).
3. Click **Launch Instance** and wait until the instance state becomes **running**.



Step 2: Connect to the Instance

1. Select your instance → Click **Connect** → Choose **EC2 Instance Connect** or SSH. Example (for SSH):

ssh -i your-key.pem ec2-user@<public-ip>



Step 3: Install CloudWatch Agent

Run the following commands inside your EC2 instance terminal:

sudo yum install amazon-cloudwatch-agent -y

```
[ec2-user@ip-172-31-34-71 ~]$ sudo yum install amazon-cloudwatch-agent -y
Amazon Linux 2023 Kernel Livepatch repository   221 kB/s |  26 kB     00:00
Dependencies resolved.
================================================================================
 Package                  Arch     Version                 Repository     Size
================================================================================
Installing:
 amazon-cloudwatch-agent  x86_64   1.300057.2-1.amzn2023   amazonlinux   135 M

Transaction Summary
================================================================================
Install  1 Package

Total download size: 135 M
Installed size: 471 M
Downloading Packages:
amazon-cloudwatch-agent-1.300057.2-1.amzn2023.x  69 MB/s | 135 MB     00:01
--------------------------------------------------------------------------------
Total                                            68 MB/s | 135 MB     00:02
Running transaction check
Transaction check succeeded.
Running transaction test
```

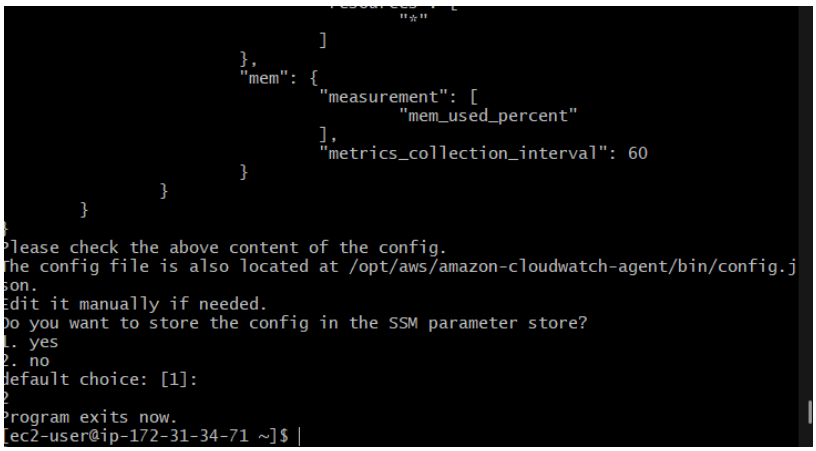
Step 4: Configure CloudWatch Agent

Start the interactive setup wizard:

sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard

When prompted:

- Choose to monitor **EC2 instance metrics**.
- Select:
    - **CPU utilization**
    - **Memory usage**
    - **Disk I/O**
- Choose to **store the config file locally** (default path: /opt/aws/amazon-cloudwatch-agent/bin/config.json).



Step 5: Start CloudWatch Agent

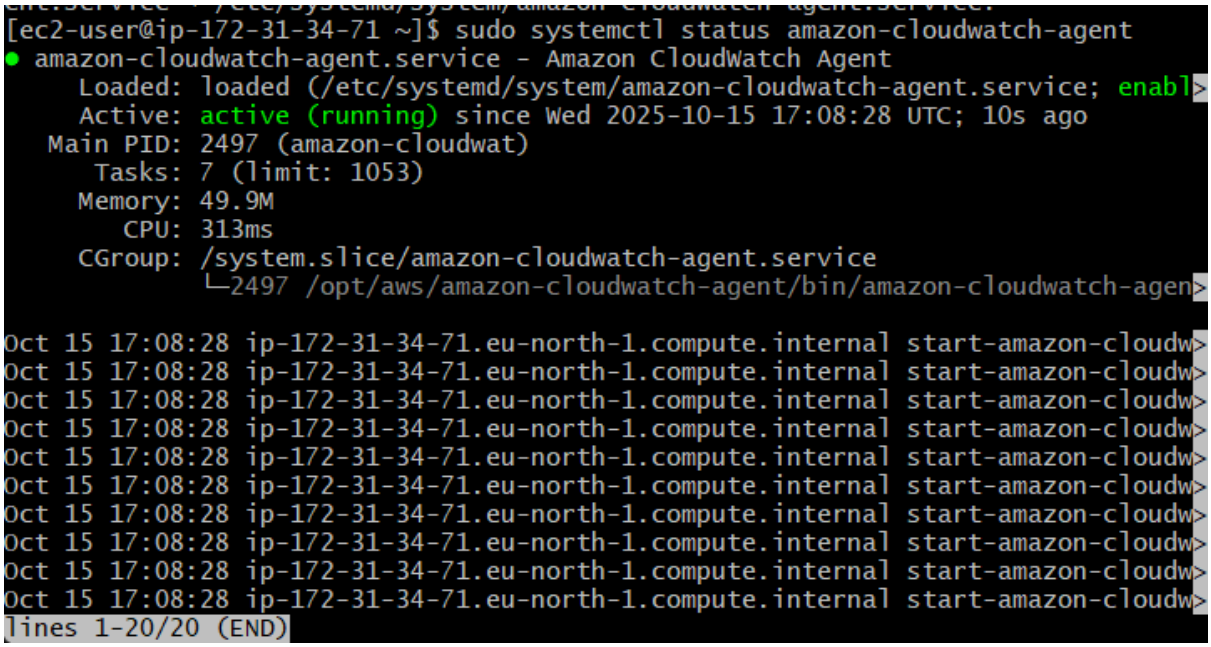
Start the agent using the configuration file:

*sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl *

-a fetch-config -m ec2 -c file:/opt/aws/amazon-cloudwatch-agent/bin/config.json -s

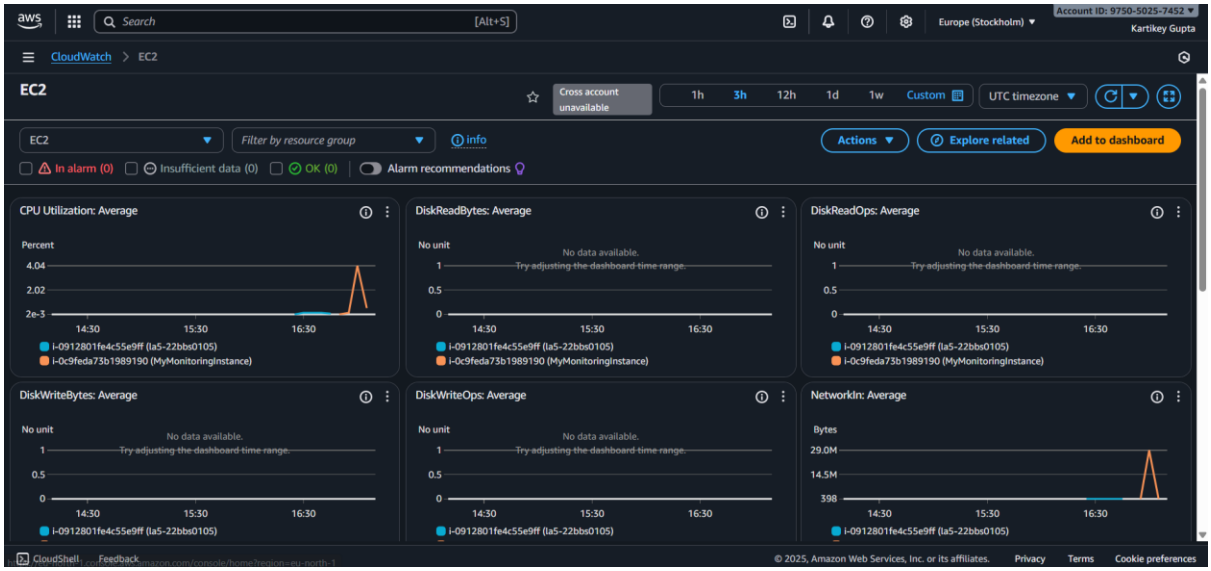
Confirm the agent is running:

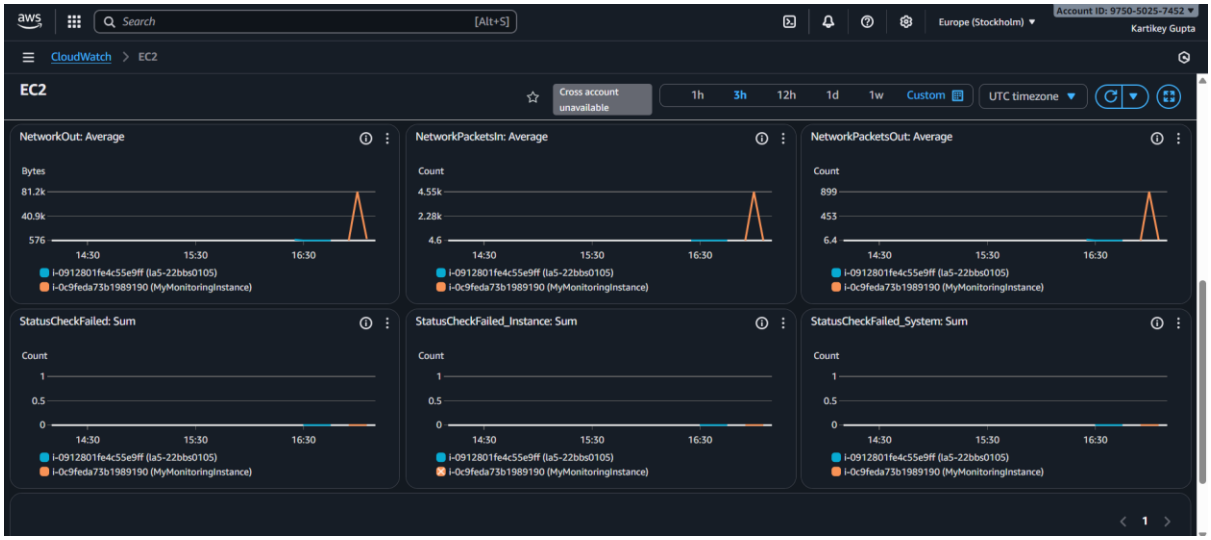
sudo systemctl status amazon-cloudwatch-agent



Step 6: Verify Metrics in CloudWatch

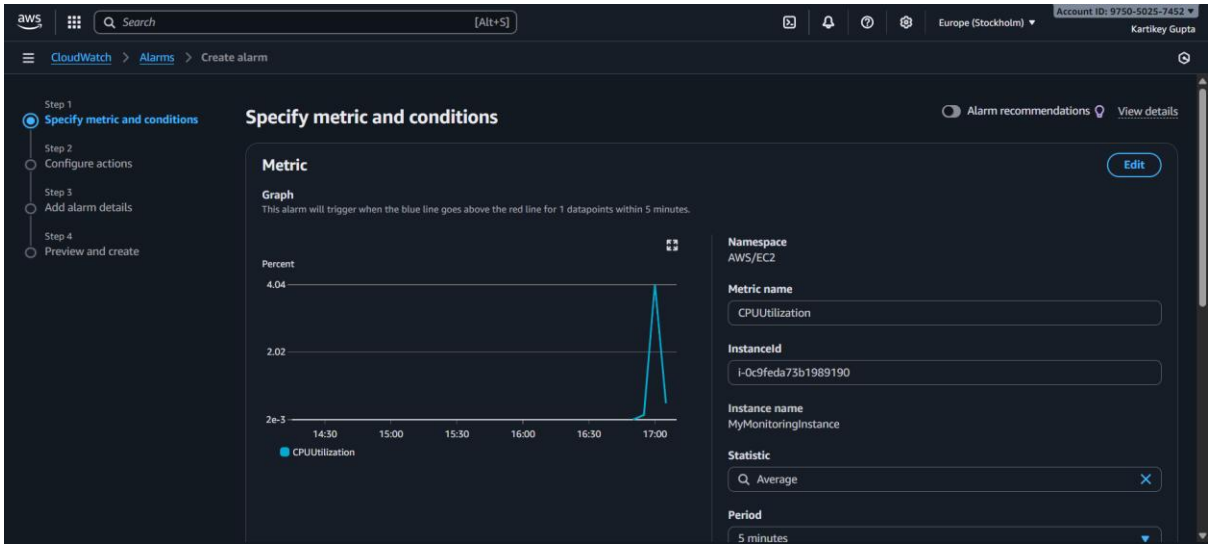
1. Go to the **CloudWatch Console** → **Metrics → All metrics → EC2**.
2. Check for metrics such as:
    - CPUUtilization
    - mem_used_percent
    - disk_write_bytes / disk_read_bytes
3. You should see your instance ID listed; click it to view real-time graphs.



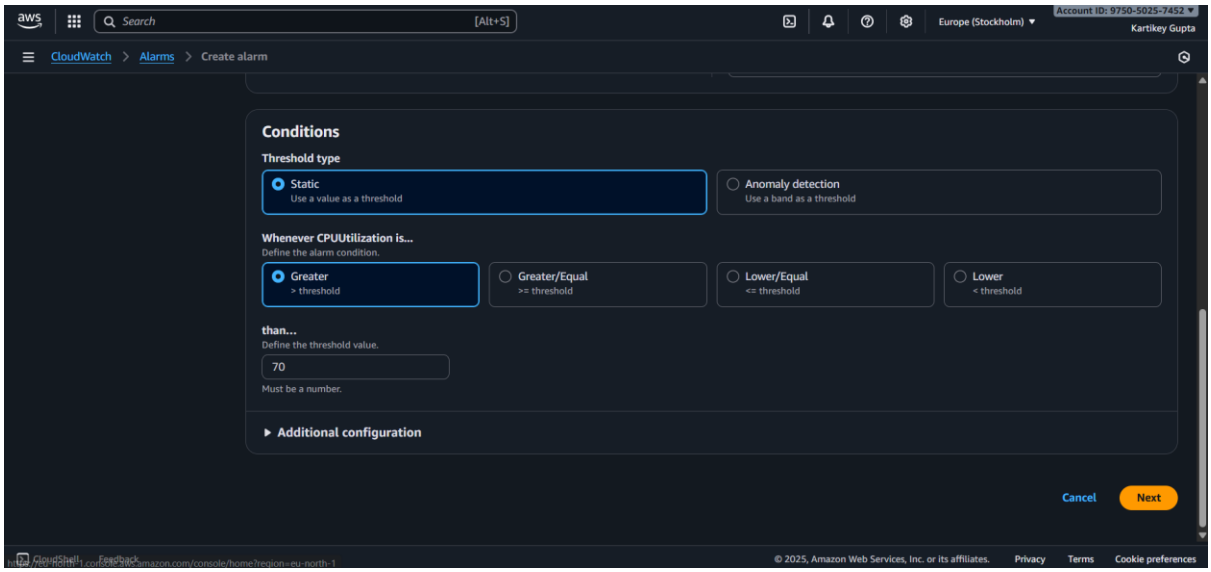


Step 7: Create CloudWatch Alarm

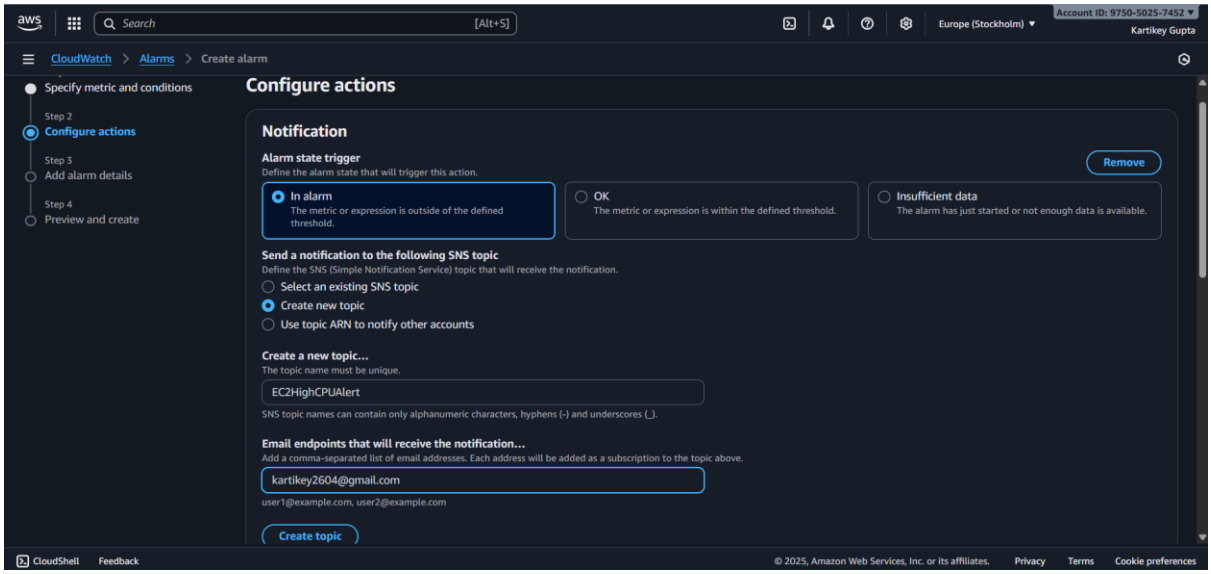
1. In the **CloudWatch Console**, go to **Alarms → All alarms → Create alarm**.
2. Choose metric:
    - **Browse metrics → EC2 → Per-Instance Metrics → CPUUtilization.**
    - Select your instance → Click **Select metric**.



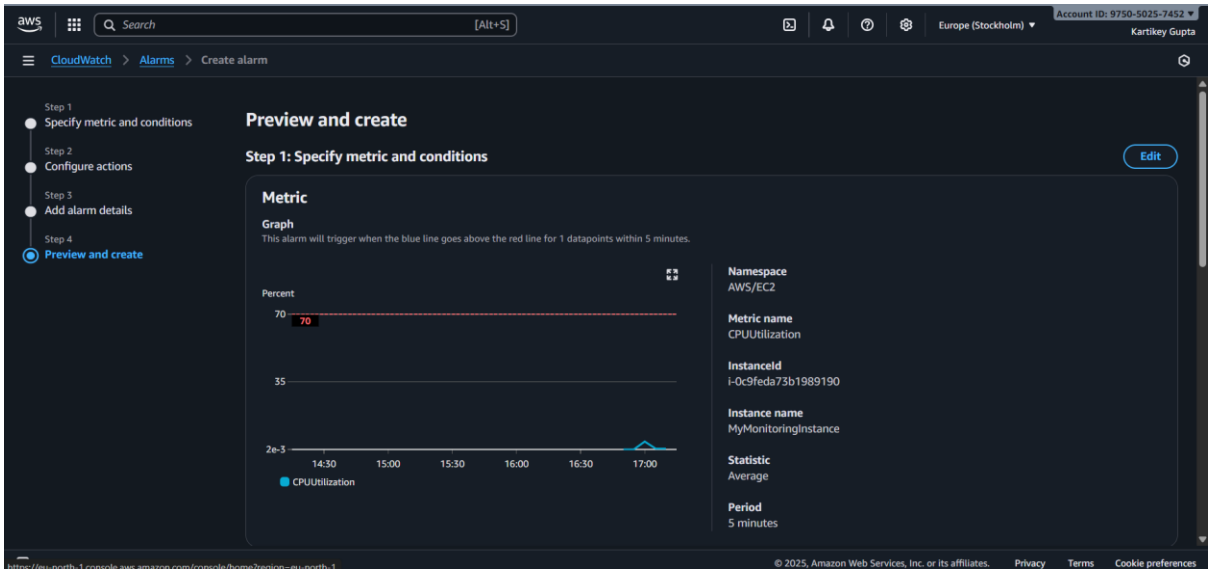
3. Set condition:
    - **Threshold type:** Static
    - **Whenever CPUUtilization > 70**
    - **For 2 consecutive 5-minute periods**



4. Create **SNS topic** for email alerts:
    - Click **Create new topic**
    - Enter topic name: EC2HighCPUAlert
    - Add your **email address** for notifications.
    - Confirm subscription from the email you receive.



5. Click **Next → Create Alarm**.

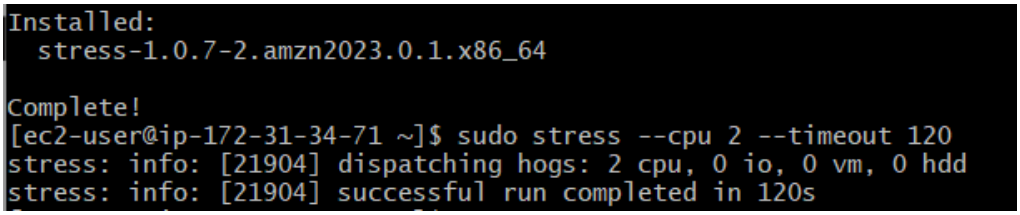


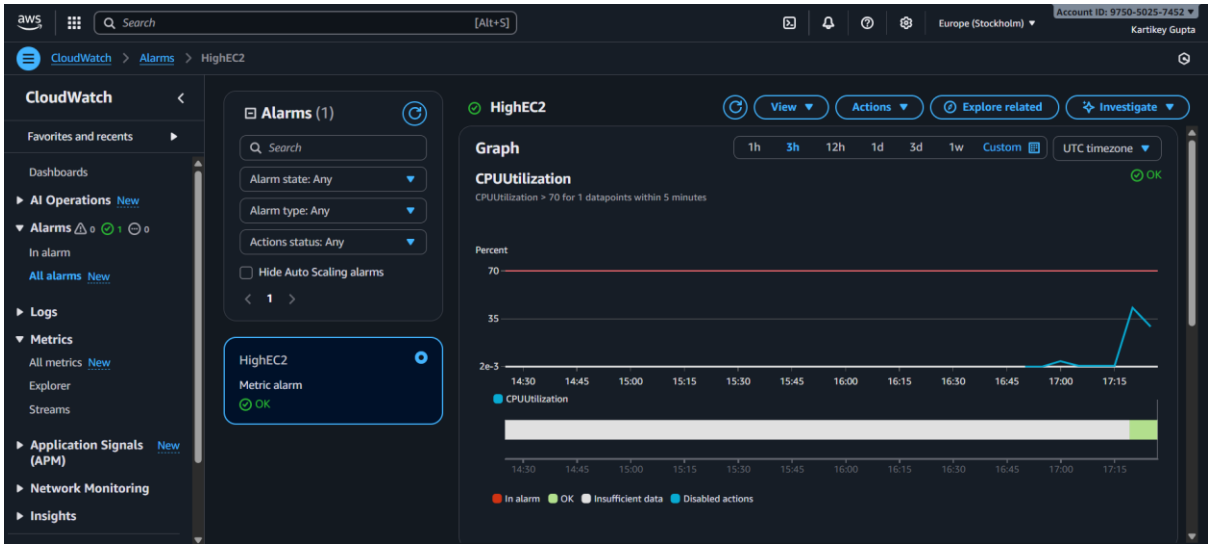
Step 8: Trigger Alarm

To test:

sudo stress --cpu 2 --timeout 120

(Install stress tool if needed: sudo yum install stress -y)
After a few minutes, CPU utilization should exceed 70%, and CloudWatch will trigger the alarm and send an email notification.

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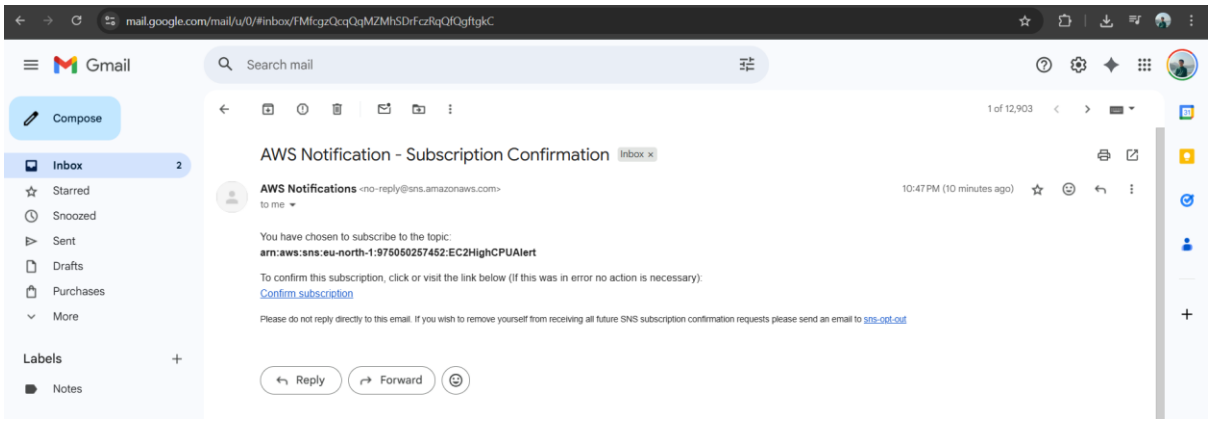
Europe (Stockholm)
Account ID: 9750-5025-7452
Kartikey Gupta
CloudWatch > Alarms > HighEC2
CloudWatch
Favorites and recents
Dashboards
AI Operations New
Alarms 0 1 0
In alarm
All alarms New
Logs
Metrics
All metrics New
Explorer
Streams
Application Signals (APM) New
Network Monitoring
Insights
Alarms (1)
Search
Alarm state: Any
Alarm type: Any
Actions status: Any
Hide Auto Scaling alarms
1
HighEC2
Metric alarm
OK
HighEC2
View
Actions
Explore related
Investigate
Graph
1h 3h 12h 1d 3d 1w Custom
UTC timezone
CPUUtilization
CPUUtilization > 70 for 1 datapoints within 5 minutes
OK
Percent
70
35
2e-3
14:30 14:45 15:00 15:15 15:30 15:45 16:00 16:15 16:30 16:45 17:00 17:15
CPUUtilization
In alarm
OK
Insufficient data
Disabled actions


mail.google.com/mail/u/0/#inbox/FMfcgzQcqQqMZMhSDrFczRqQfQgftgkC
Gmail
Search mail
Compose
Inbox 2
Starred
Snoozed
Sent
Drafts
Purchases
More
Labels
Notes
1 of 12,903
AWS Notification - Subscription Confirmation
Inbox
AWS Notifications <no-reply@sns.amazonaws.com>
to me
10:47 PM (10 minutes ago)
You have chosen to subscribe to the topic:
arn:aws:sns:eu-north-1:975050257452:EC2HighCPUAlert
To confirm this subscription, click or visit the link below (If this was in error no action is necessary):
Confirm subscription
Please do not reply directly to this email. If you wish to remove yourself from receiving all future SNS subscription confirmation requests please send an email to sns-opt-out
Reply
Forward